D-ANMOL IT SOLUTIONS

Artificial Intelligence & Machine

Learning

6 Months | Advanced Professional Track

A step-by-step practical learning roadmap with explained tools, applied examples, project milestones and career preparation.

01 WHAT CAN YOU BUILD AFTER THIS COURSE?

- Student Performance Predictor: Train a model to estimate student scores from attendance and study patterns.
- Spam Email Classification: Classify text messages as spam or legitimate with a tested pipeline.

Possible career directions

ML trainee, junior AI/ML developer, data analyst and applied AI project intern. Roles are illustrative and depend on skills, experience and hiring criteria.

Your learning journey

From learning to practical delivery

Learn concepts	Practise skills	Build projects	Show portfolio

02 STEP-BY-STEP COURSE SYLLABUS

Modules are ordered from foundations to application. Each includes what you learn, why it matters and a practical classroom activity.

STEP 01 | AI, ML & Deep Learning Overview

What you will study:

- AI history: understand the concept, practise it with a guided example, and apply it in a relevant task.
- narrow AI: understand the concept, practise it with a guided example, and apply it in a relevant task.
- ML types: understand the concept, practise it with a guided example, and apply it in a relevant task.
- deep learning: understand the concept, practise it with a guided example, and apply it in a relevant task.
- use cases: understand the concept, practise it with a guided example, and apply it in a relevant task.
- workflow and responsible AI: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: It provides a foundation for later modules and real project implementation.

Hands-on example: Map a real-world problem to a suitable AI approach.

STEP 02 | Python for AI

What you will study:

• Variables: understand the concept, practise it with a guided example, and apply it in a relevant task.

• data types: understand the concept, practise it with a guided example, and apply it in a relevant task.

• conditions: understand the concept, practise it with a guided example, and apply it in a relevant task.

• loops: understand the concept, practise it with a guided example, and apply it in a relevant task.

• functions: understand the concept, practise it with a guided example, and apply it in a relevant task.

• collections: understand the concept, practise it with a guided example, and apply it in a relevant task.

• modules and file handling: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: It provides a foundation for later modules and real project implementation.

Hands-on example: Write reusable Python programs for data tasks. D-ANMOL IT SOLUTIONS | COURSE CURRICULUM Page 2

STEP 03 | NumPy & Pandas

What you will study:

• Arrays: understand the concept, practise it with a guided example, and apply it in a relevant task.

• indexing: understand the concept, practise it with a guided example, and apply it in a relevant task.

• DataFrames: understand the concept, practise it with a guided example, and apply it in a relevant task.

• importing data: understand the concept, practise it with a guided example, and apply it in a relevant task.

• filtering: understand the concept, practise it with a guided example, and apply it in a relevant task.

• missing values and transformations: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: It provides a foundation for later modules and real project implementation.

Hands-on example: Clean a raw dataset and produce an analysis-ready table.

STEP 04 | Maths & Statistics for ML

What you will study:

• Descriptive statistics: understand the concept, practise it with a guided example, and apply it in a relevant task.

• probability: understand the concept, practise it with a guided example, and apply it in a relevant task.

• distributions: understand the concept, practise it with a guided example, and apply it in a relevant task.

• vectors: understand the concept, practise it with a guided example, and apply it in a relevant task.

• matrices and correlation: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Summarise a dataset and interpret statistical relationships.

STEP 05 | Data Visualisation & EDA

What you will study:

• Matplotlib: understand the concept, practise it with a guided example, and apply it in a relevant task.

• Seaborn: understand the concept, practise it with a guided example, and apply it in a relevant task.

• distributions: understand the concept, practise it with a guided example, and apply it in a relevant task.

• outliers: understand the concept, practise it with a guided example, and apply it in a relevant task.

• relationships and exploratory analysis: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Create an EDA notebook with visual findings.

STEP 06 | ML Workflow & Feature Engineering

What you will study:

• Problem framing: understand the concept, practise it with a guided example, and apply it in a relevant task.

• labels: understand the concept, practise it with a guided example, and apply it in a relevant task.

• features: understand the concept, practise it with a guided example, and apply it in a relevant task.

• train/test split: understand the concept, practise it with a guided example, and apply it in a relevant task.

• encoding: understand the concept, practise it with a guided example, and apply it in a relevant task.

• scaling and leakage prevention: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Prepare a reusable model-ready dataset.

STEP 07 | Supervised Learning: Regression

What you will study:

• Linear regression: understand the concept, practise it with a guided example, and apply it in a relevant task.

• multiple regression: understand the concept, practise it with a guided example, and apply it in a relevant task.

• assumptions: understand the concept, practise it with a guided example, and apply it in a relevant task.

• residuals and regression metrics: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Build a house-price or sales prediction model.

STEP 08 | Supervised Learning: Classification

What you will study:

• Logistic regression: understand the concept, practise it with a guided example, and apply it in a relevant task.

• KNN: understand the concept, practise it with a guided example, and apply it in a relevant task.

• decision trees: understand the concept, practise it with a guided example, and apply it in a relevant task.

• confusion matrix: understand the concept, practise it with a guided example, and apply it in a relevant task.

• precision: understand the concept, practise it with a guided example, and apply it in a relevant task.

• recall and F1: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Build a spam or customer-churn classifier.

STEP 09 | Ensemble Models & Model Selection

What you will study:

• Random forest: understand the concept, practise it with a guided example, and apply it in a relevant task.

• boosting concepts: understand the concept, practise it with a guided example, and apply it in a relevant task.

• cross-validation and model comparison: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Compare multiple models and justify the selected model.

STEP 10 | Unsupervised Learning

What you will study:

• K-Means: understand the concept, practise it with a guided example, and apply it in a relevant task.

• hierarchical clustering: understand the concept, practise it with a guided example, and apply it in a relevant task.

• PCA concepts and segmentation use cases: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Segment customers or products using clustering.

STEP 11 | Model Tuning & Interpretation

What you will study:

• Hyperparameters: understand the concept, practise it with a guided example, and apply it in a relevant task.

• grid/random search: understand the concept, practise it with a guided example, and apply it in a relevant task.

• feature importance: understand the concept, practise it with a guided example, and apply it in a relevant task.

• explainability basics and limitations: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Tune a model and document its strengths and weaknesses. D-ANMOL IT SOLUTIONS | COURSE CURRICULUM Page 3

STEP 12 | Capstone & Responsible AI

What you will study:

• End-to-end workflow: understand the concept, practise it with a guided example, and apply it in a relevant task.

• bias: understand the concept, practise it with a guided example, and apply it in a relevant task.

• privacy: understand the concept, practise it with a guided example, and apply it in a relevant task.

• reproducibility: understand the concept, practise it with a guided example, and apply it in a relevant task.

• documentation and presentation: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Present a complete ML project with a report and demo.

03 TOOLS, LIBRARIES & THEIR USES

This section explains where each tool or library fits in a practical project. Some advanced tools may be introduced as demonstrations in shorter tracks.

Tool / library	What it is used for
NumPy	Perform fast numerical operations on arrays.
Pandas	Load, clean and transform tabular datasets.
Matplotlib / Seaborn	Visualize distributions, trends and model results.
Scikit-learn	Preprocess features, train models and evaluate predictions.
Jupyter Notebook	Run experiments and document analysis step by step.

04 PROJECT-BASED LEARNING

Included: 2 project(s). Every project includes planning, implementation, testing and a presentation of results.

LIVE PROJECT 1: Spam Email Classification

What you will build: Classify text messages as spam or legitimate with a tested pipeline.

Where tools are used: Scikit-learn TF-IDF vectorizer and classifier; Pandas for labels.

Final deliverable: Classifier, confusion matrix and test samples.

Project workflow

Input / brief	Build & test	Review results	Present output

Live project means an applied, end-to-end project based on a realistic business scenario or suitable dataset; access to an external client or production deployment is not implied.

LIVE PROJECT 2: Customer Segmentation Model

What you will build: Group shoppers based on purchase behaviour to suggest marketing segments.

Where tools are used: Pandas, StandardScaler, KMeans and Seaborn.

Final deliverable: Segment profiles, visualizations and documented limitations.

Project workflow

Input / brief	Build & test	Review results	Present output

Live project means an applied, end-to-end project based on a realistic business scenario or suitable dataset; access to an external client or production deployment is not implied.

05 TRAINING PLAN & ASSESSMENTS

Suggested duration: 6 Months (approximately 24 weeks). Exact session hours and timetable should be confirmed with the institute.

- Month 1–2 — Foundations, tools and regular practice
- Month 3–4 — Intermediate/advanced topics and first live project
- Month 5 — Second live project, documentation and portfolio
- Month 6 — Interview classes, mock interviews and capstone presentation

Evaluation: practical exercises, module checkpoints, code/document review where applicable, project demonstration and final feedback.

06 INCLUDED INTERVIEW CLASSES

- Resume and portfolio presentation for the selected course.
- Technical and practical interview questions based on course modules.
- Mock interviews, communication practice and feedback.
- Explaining project architecture, decisions, challenges and results.

FREE SUPPORT INCLUDED WITH EVERY COURSE

1 YEAR DOUBT SUPPORT + INTERVIEW PREPARATION — FREE OF COST

- One year of doubt-clearing support related to the enrolled course.
- Interview preparation support at no additional course fee.
- Guidance on explaining course projects and strengthening fundamental concepts.
- Availability, format and scheduling of support sessions to be communicated by the institute.

Important: This brochure presents the requested training offer. The institute should confirm the exact support policy, availability, start date and any applicable conditions before publishing. Interview preparation is not a job or placement guarantee.

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